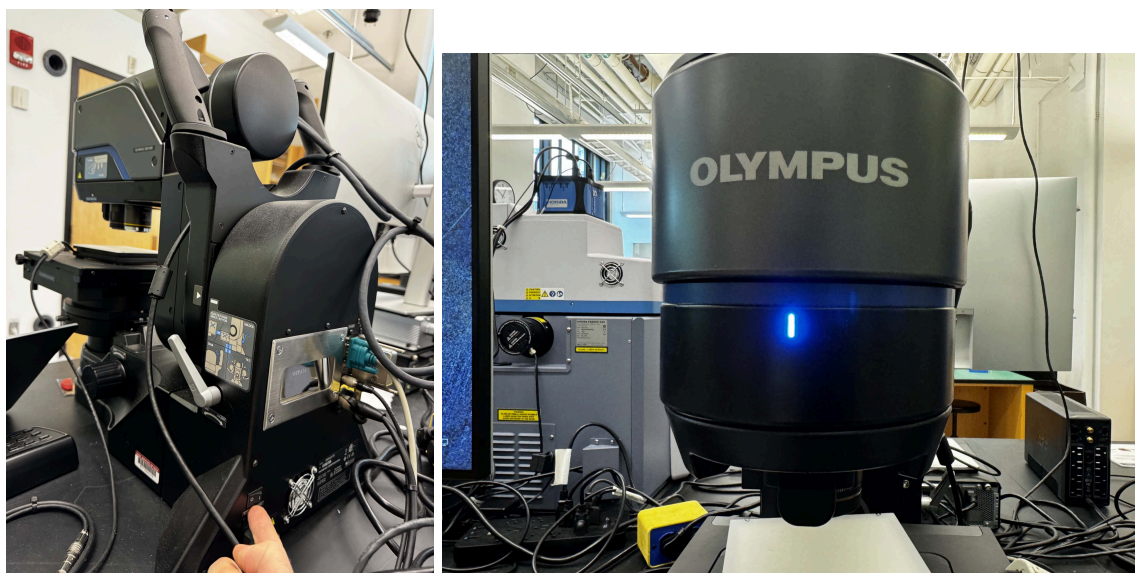


## 3.000 - Optical Microscope Station

In this rotation, you compare different coffee filters using the optical microscope. You will measure and compare the hole size of a cloth, paper, and metal filter. *You will need to measure three hole sizes per filter. You will submit these nine values and their averages as a Canvas document once this rotation is complete.*

### Instructions:

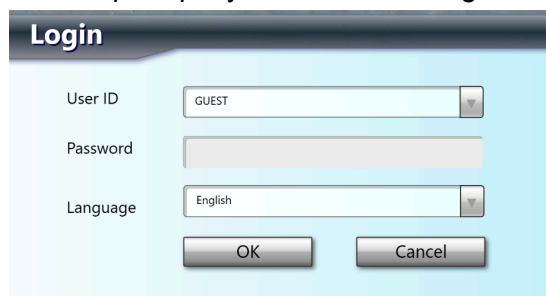
1. Remove the dust cover from the optical microscope.
2. Turn the microscope on using the switch on the back. The front light should appear blue once the microscope is on.



3. Log onto the computer using your MIT kerberos.
4. Clear the microscope stage of any samples or other materials



5. On the computer, click the logo for “DSX”
6. When it prompts you, click OK to log on as a Guest.



The login dialog box has a title bar that says "Login". It contains three input fields: "User ID" with a dropdown menu showing "GUEST", "Password" with an empty text box, and "Language" with a dropdown menu showing "English". At the bottom are two buttons: "OK" and "Cancel".

7. Click “OK” to acknowledge that it is safe for the stage and head to move.
8. Use the manual focusing knob to lower the microscope stage.
9. Place a filter on the microscope stage.



10. Focus on the filter (three options):
  - a. Use the manual focus knob
  - b. Use the digital focus arrows shown in the image.
  - c. Click “AF” to Autofocus.



11. Use the joystick control to move the stage left/right/forward/backwards.

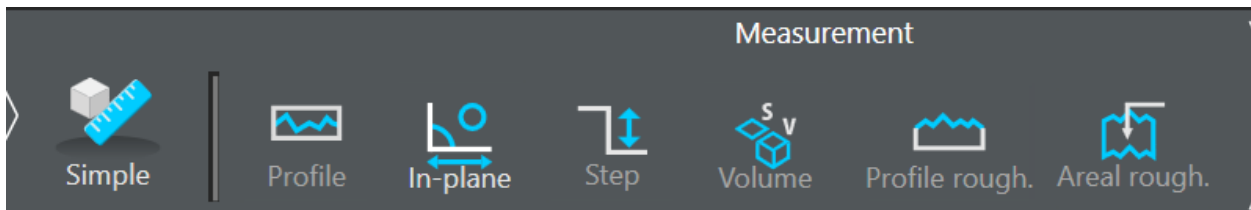


12. Zoom into a group of filter pores using the Zoom slider.
13. Take a picture by clicking “2D Acquisition”

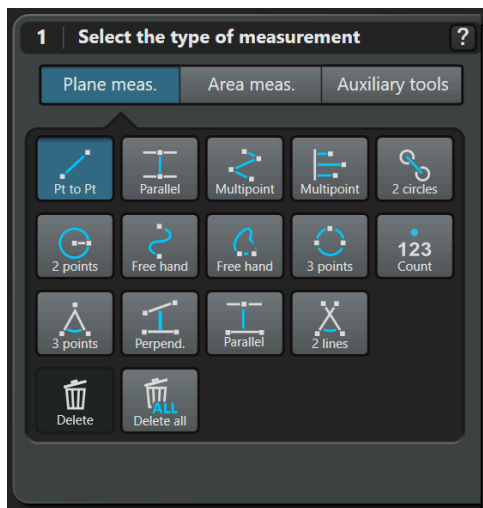


14. You do not need to save your image. The analysis application for the microscope will automatically open.

15. Click on “In-plane” under the measurement tab.



16. Depending on the filter, use different measurement methods.
  - a. For the metal filter, click “3 points” then click along the edge of a pore to measure the radius of the pore.
  - b. For the paper and the cloth filters, the pores will be less clear. Click “Pt to Pt” to measure the distance between two points.



17. **Complete three measurements of pore size for each filter.**
18. Once you are finished, remove your sample from the stage.
19. To save any images, you can email them to yourself, move it to a USB stick or move it to the common Breakerspace Dropbox. Visit <https://breakerspace.mit.edu/resources.html> to learn how to sign up to the Breakerspace Dropbox